

1. TALK 1. WHY DO FORMAL GROUP LAWS ARISE?

Two main references are: Lecture 1 [1], [2] for details about complex characteristic classes, the classifying space $BU(n)$ and complex K -theory.

- Define Chern characteristic classes, discuss operations on the set of vector bundles.
- State the existence of the classifying space $BU(n)$ and show a sketch of the computation of its cohomology
- Define complex-oriented and even cohomology theory, explain their importance
- Work out two main examples: complex K -theory and integral cohomology, explain how formal group laws arise in these two examples
- Define a formal group law

2. TALK 2. EQUIVALENCES OF FORMAL GROUPS

- Briefly outline the picture from differential geometry: the beginning of the Lecture 2 [1] and I.'s notes or any other source.
- Introduce étale sheaves as geometric setting, follow Lecture 2 [1], skip the part about the category of elements: [Definition 2.1], [Remark 2.2], some examples, [Definition 2.16] and below.
- Define formal groups and show how formal group laws produce formal groups, follow Lecture 3 [1]: [Definition 3.1], [Proposition 3.5], *only if time permits*, [Definition 3.7], [Lemma 3.9], [Definition 3.14]
- Work out the example of complex-oriented cohomology theory and KU if time permits: [Example 3.17] and [Remark 3.18] [1], respectively. Show how $\mathbb{G}_m$ and $\mathbb{G}_a$ arise as formal groups associated to KU and $H\mathbb{Z}$ respectively.
- Sketch that over $\mathbb{Q}$ all formal group laws are equivalent: [Theorem 5.13] [1] using an intuition from differential geometry and taking some statements on faith (such as [Lemma 5.2] and [Lemma 5.11]).

3. TALK 3. MODULI OF FORMAL GROUPS

All references are from [1].

- Representability of functors, show moduli problems (for FG and FGL) representable, and what they are represented by, follow Piotr
- Define the functor $\mathcal{F}gl$ and prove that the Lazard ring which represents it as in [Lemma 6.1, Definition 6.2]. State Lazard's theorem without proof [Theorem 6.8] and explain how the proof over $\mathbb{Q}$ follows from [Theorem 5.13]: [Corollary 6.10]. and deduce that formal group laws can always be lifted along surjective homomorphisms [Corollary 6.9].
- Define the moduli stack of formal groups $\mathcal{M}_{fg}$ following [Definition 12.1]. State and prove [Prop.12.2, Prop 12.3]. Define sheaves with values in $\mathcal{A}nimae$ [Definition 8.1]

4. TALK 4. COMPLEX ORIENTABILITY & MU IS THE UNIVERSAL COMPLEX ORIENTED THEORY

All references are from [1].

There is an ∞ -categorical Remark 9.7., can we try to present material in that spirit?

- In this talk, we will introduce the cohomology theory MU of complex bordism and show that MU is universal among complex-oriented cohomology theories.
- Briefly recall what an E_∞ -ring spectrum is.
- Define MU as an E_∞ -ring spectrum. State and prove [Theorem 9.14.].
- State Quillen's theorem [Theorem 9.15, Theorem 11.8] + (Remark 9.11) and explain its consequence following [Theorem 11.10] and the paragraphs before and after it if time permits.

5. TALK 5. FORMAL GROUPS CLASSIFY CERTAIN COHOMOLOGY THEORIES

- State and prove [Proposition 12.8], [Proposition 12.2] + [Proposition 12.3] too.
- Hence deduce that we have a map

$$\mathrm{Sp} \longrightarrow \mathcal{QCoh}(\mathcal{M}_{\mathrm{fg}})$$

following [Rognes, Definition 11.5.2], [Theorem 11.5.1] as well.

6. TALK 6. DEFORMATION THEORY AND EXAMPLES

7. TALK 6. FROBENIUS AND HEIGHTS

- Define the Frobenius automorphism [Definition 13.1] [1]
- Define what it means to have height at least n [Definition 13.8] [1] + explain its' connection with a "usual" definition (see for example Lecture 12 [3]) + [Example 13.16] [1]
- Explain that not all formal groups have a height [Definition 13.11] and below, [Corollary 13.13]
- Define v_n 's [Definition 13.17] [1] + [Corollary 13.19] [1]
- Introduce the height filtration on the moduli stack of formal groups [Proposition 14.1] + [Proposition 14.3] [1]

8. TALK 7. HOW TO GO FROM FORMAL GROUPS TO COHOMOLOGY THEORIES: FORMS OF K-THEORY **TO BE UPDATED**

- Show how to construct cohomology theories from formal groups: Construction 19.1 [1]
- Formulate the Landweber Exact Functor Theorem: Theorem 19.5 [1]
- Show how deformation theory is used to construct Landweber exact sequences: Corollary 18.19, Lemma 20.21, Definition 20.22, Theorem 20.23

9. TALK 8. INTRO TO RING SPECTRA AND COHOMOLOGY THEORIES E_∞ , E_1

10. TALK 9. SPECTRA ARE QUASICOHERENT SHEAVES ON THE MODULI STACK OF FORMAL GROUPS

Supposed to be covered: (Relation between Sp and $\mathcal{QCoh}(\mathcal{M}_{\mathrm{fg}})$) Explain how to extend the action in $\mathrm{inf}\text{-cat.}$ terms, $\mathrm{Spec} S$ and $\mathrm{Spec} \mathrm{Laz}$ iso, mention that by a module constructed above we obtain a sheaf on $\mathrm{Spec} L$, upgrade it to define map $\mathrm{Sp} \longrightarrow \mathcal{QCoh}(\mathcal{M}_{\mathrm{fg}})$, state nilp theorem and what it tells us about the map.

11. TALK 10. SPECTRAL PRIMES: MORAVA E-THEORY AND MORAVA K-THEORY

Supposed to be covered: Constructing Morava E -theory and Showing its E_∞ , Morava K -theory as E_1 ring. Using *LEFT*, introduce Lurie's obstruction theory theorem (faithful flatness)

12. TALK 11. FORMS OF K-THEORY: GETTING TO KNOW THE FIRST ELLIPTIC COHOMOLOGY

Supposed to be covered: Hands on $E(1) = KU_p^\wedge$. Description of $L_{K(1)}S$. Talk of illustrating properties about it that we've covered in each talk.

13. TALK 12. CHROMATIC FRACTURE SQUARE

Supposed to be covered: The Chromatic Fracture Square. Bousfield localisation, $L_{E(n)}$ is a smashing localisation. Height filtration.

14. TALK 13. E-THEORY IS A GALOIS COVER OF THE $K(H)$ -LOCAL SPHERE

Supposed to be covered: *The Devinatz–Hopkins Theorem*

REFERENCES

- [1] P. Pstragowski, *Finite Height Chromatic Homotopy Theory*, Lecture notes for Math 252Y, Harvard University, Spring 2021.
- [2] J. Rognes, *Chromatic Homotopy Theory*, Algebraic Topology III, MAT4580/MAT9580, University of Oslo, 2023.
- [3] J. Lurie, *Chromatic Homotopy Theory*, lecture notes for Math 252, Harvard University.